

# SCANNING NETWORKS LAB REPORT

| BACKGROUND                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                        |
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| Course:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | CIS 170 - Hands-On Ethical Hacking and Network Defense |
| Term:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Spring 2026                                            |
| Instructor:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Ray Lampano, Jnr.                                      |
| Section Number:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 15480/15481                                            |
| Student Name:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Donaven Bruce                                          |
| Student ID:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | [Redacted]                                             |
| Submission Platform:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Canvas                                                 |
| Report Date:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 2026-01-25                                             |
| Report Version:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 1.0                                                    |
| Lab Title:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Scanning Networks                                      |
| Module Reference:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Scanning Networks                                      |
| Lab Type:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Live Virtual Machine Lab                               |
| Date Completed:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 2026-01-25                                             |
| Time Spent:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Approximately 1 hour                                   |
| ENVIRONMENT                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                        |
| <b>Platforms and Systems:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                        |
| <ul style="list-style-type: none"><li>- Primary environment: Practice Labs / Live Virtual Machine Labs controlled sandbox.</li><li>- Primary systems available in the topology: PLABDC01 (Windows Server 2019 - Domain Controller), PLABDM01 (Windows Server 2019 - Domain Member), PLABWIN10 (Windows 10 - Domain Member), and PLABKALI01 (Kali 2019.2 - Linux Kali Workstation).</li><li>- The work centered mainly on PLABWIN10 for Microsoft-side tools (ID Serve, Zenmap, CurrPorts, MyLanViewer) and PLABKALI01 for Linux-side tools (Nmap, Netdiscover, fping, Netcat, Hping3, Dmitry, msfconsole), with scans targeting the full lab topology (e.g., 192.168.0.0/24 range and specific hosts).</li><li>- Remote access to the virtual machines was provided through the course platform.</li></ul> |                                                        |
| <b>Context and Constraints:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                        |
| <ul style="list-style-type: none"><li>- The module focused on network scanning concepts, host discovery, port scanning, service/version enumeration, OS fingerprinting, banner grabbing, and defensive detection of scanning activity.</li><li>- Practice Labs flagged the module as legacy content using deprecated software, so behavior could vary and some steps were best interpreted as guided sandbox demonstrations.</li><li>- Windows Updates were disabled in the Practice Labs environment, so software behavior reflected a controlled legacy setup rather than a current hardened enterprise baseline.</li><li>- The module aligned most closely with exam objective 5.2 (Information Security Assessment Methodologies).</li></ul>                                                           |                                                        |
| <b>Authorization and Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                        |
| <ul style="list-style-type: none"><li>- All documented activity took place only inside approved academic systems and isolated practice-lab sandboxes.</li><li>- No public IP space, real organizations, or unauthorized targets were involved.</li><li>- No production infrastructure, third-party systems, or live services were contacted.</li><li>- The actions documented here were performed for defensive understanding, reconnaissance analysis, and ethical hacking education within the course environment.</li><li>- Restricted material, live credentials, and personal data are omitted or redacted from this report.</li></ul>                                                                                                                                                                |                                                        |
| <b>Work Objectives:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                        |
| <ul style="list-style-type: none"><li>- Review the basic components of network scanning, host discovery techniques, port-scanning types (TCP connect, SYN, UDP, etc.), service enumeration, OS fingerprinting, and banner grabbing.</li><li>- Use Microsoft-side tools (ID Serve, Zenmap, CurrPorts, MyLanViewer) on PLABWIN10 to perform host discovery, port scans, banner grabbing, and connection monitoring.</li><li>- Use Linux-side tools (Nmap with multiple scan types, Netdiscover, fping, Netcat, Hping3, Dmitry, msfconsole) on PLABKALI01 to perform advanced scans and analyze results across the lab topology.</li></ul>                                                                                                                                                                    |                                                        |
| <b>Scanning Networks Concepts and Defensive Framing:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                        |
| <ul style="list-style-type: none"><li>- The lab identified three core phases of network scanning: host discovery, port scanning, and service/OS enumeration.</li><li>- Scan types reviewed included ping sweeps, TCP connect scans, SYN stealth scans, UDP scans, and version detection, with evasion techniques (timing, fragmentation) also covered.</li><li>- Tools demonstrated both GUI (Zenmap, MyLanViewer) and CLI approaches (Nmap, Hping3), showing how scanning reveals live hosts, open ports, running services, and OS details.</li><li>- Additional methods reviewed included banner grabbing and connection monitoring, reinforcing that scanning is the foundation of reconnaissance and targets network visibility rather than only technology exploits.</li></ul>                        |                                                        |

## TECHNICAL ANALYSIS

### Microsoft Network Scanning Tools on PLABWIN10:

- On PLABWIN10, ID Serve was launched from the desktop shortcut (via Intranet Hacking Tools), the Server Query tab was selected, `http://intranet` was entered, and Query The Server was clicked to retrieve detailed web server banners and configuration data.
- Zenmap (Nmap GUI) was started from the desktop, the Target field set to `192.168.0.1`, the Intense scan profile selected, and Scan clicked to produce host discovery, port, service, and topology results (Nmap Output, Ports/Hosts, Topology, and Host Details tabs).
- CurrPorts was launched from the extracted cports folder and used to monitor active TCP/UDP connections in real time, showing process names, PIDs, local/remote addresses, ports, and connection states; an attempt was made to close a selected TCP connection (requiring elevated privileges).
- MyLanViewer was installed and launched, performing an automatic subnet scan (`192.168.0.0/24`); PLABWIN10 was expanded to view MAC, IP, and system details, and Ping/Traceroute was run against `192.168.0.250`.

### Linux Network Scanning Tools on PLABKALI01:

- On PLABKALI01, the terminal was opened and Hping3 was used for ICMP discovery (`hping3 192.168.0.1 -icmp -c 5`), TCP SYN port scanning (`hping3 intranet -scan 80,22,443,1-80,1-65535 -S`), and open-port identification (`hping3 -8 0-100 -S`).
- Dmitry was executed with `dmitry -p intranet` (TCP port scan) and `dmitry -p intranet -o plab` (output saved to `plab.txt`, then viewed with `cat plab.txt`).
- Netcat was used for port scanning with `nc -nvz 192.168.0.1 88` (and other ports).
- Netdiscover performed active ARP discovery with `netdiscover -r 192.168.0.0/24` and passive stealth scanning with `netdiscover -p`.
- Nmap was used for stealth scanning: `nmap -sS 192.168.0.1`, `nmap -sA 192.168.0.0/24`, `nmap -sN 192.168.0.1-4`, `nmap -sF 192.168.0.1-4`, `nmap -sX 192.168.0.1-3`, and timing-controlled scan `nmap 192.168.0.2 -T 5`.
- fping performed host discovery with `fping 192.168.0.1`, `fping -g 192.168.0.1 192.168.0.2`, and full subnet scan `fping -g 192.168.0.0/24`.
- msfconsole was launched, the auxiliary/scanner/portscan/syn module loaded, RHOSTS, THREADS, and PORTS set, and run executed against single hosts, port ranges, and multiple hosts (`192.168.0.1-3`).

## KEY COMMANDS, TOOLS, AND ARTIFACTS BY SYSTEM

### PLABWIN10 — Microsoft Scanning Tools:

ID Serve (Server Query tab)  
 http://intranet (Query The Server)  
 Zenmap  
 Target: 192.168.0.1  
 Profile: Intense scan  
 Scan  
 CurrPorts (monitor TCP/UDP)  
 MyLanViewer (subnet scan)  
 Ping/Traceroute To Host: 192.168.0.250

### PLABKALI01 — Linux Scanning Tools / Nmap Workflow:

```
hping3 192.168.0.1 -icmp -c 5
hping3 intranet -scan 80 -S
hping3 intranet -scan 22,80,443 -S
hping3 192.168.0.1 -scan 1-80 -S
hping3 192.168.0.1 -scan 1-65535 -S
hping3 -8 0-100 -S 192.168.0.1
dmitry -p intranet
dmitry -p intranet -o plab
cat plab.txt
nc -nvz 192.168.0.1 88
netdiscover -r 192.168.0.0/24
netdiscover -p
nmap -sS 192.168.0.1
nmap -sA 192.168.0.0/24
nmap -sN 192.168.0.1-4
nmap -sF 192.168.0.1-4
nmap -sX 192.168.0.1-3
nmap 192.168.0.2 -T 5
fping 192.168.0.1
fping -g 192.168.0.0/24
msfconsole
use auxiliary/scanner/portscan/syn
set RHOSTS 192.168.0.1
set THREADS 25
set PORTS 53
set PORTS 1-100
set RHOSTS 192.168.0.1-3
run
```

### Primary Tools and Artifacts Observed:

- ID Serve banner output, Zenmap Nmap Output / Ports / Hosts / Topology / Host Details tabs, CurrPorts socket list, MyLanViewer network map with MAC/IP details, Hping3 SYN/ACK responses, Dmitry TCP port scan and plab.txt output, Netcat verbose port results, Netdiscover ARP table (active/passive), Nmap stealth scan tables (SYN/ACK/FIN/NULL/Xmas), fping live host list, and msfconsole syn scanner results.
- The most important evidence of scanning activity in the lab was the Nmap host-discovery output listing live systems, detailed port/service/OS tables from multiple scan types, banner-grab results from ID Serve, and real-time connection monitoring in CurrPorts and MyLanViewer.

## DEFENSIVE MAPPING AND SUMMARY

### Technique Summary:

- The completed work covered network-scanning concepts, host discovery (ping sweeps, ARP), multiple port-scanning techniques across Microsoft and Linux tools, service/version detection, OS fingerprinting, banner grabbing, and network mapping.
- The recurring theme across the module was that reconnaissance via scanning provides the attacker with a detailed map of live systems, open ports, and running services before any exploitation occurs.

### MITRE ATT&CK Mapping:

- Reconnaissance: T1595 Active Scanning - the Nmap SYN/port scans and fping/Netdiscover host discovery directly mapped to active network probing.
- Reconnaissance: T1595.001 Active Scanning: Scanning IP Blocks - the full subnet scans (192.168.0.0/24) aligned with broad IP-block reconnaissance.
- Discovery: T1046 Network Service Discovery - service/version detection and banner grabbing demonstrated discovery of running services and software versions.
- Execution / Persistence context: the detailed host/port/service data collected can feed into subsequent exploitation or lateral movement phases.

### Detection Opportunities:

- Unusual volumes of ICMP echo requests, SYN packets without completion, or rapid port probes are strong indicators of scanning activity.
- IDS/IPS signatures for Nmap patterns, Hping3 crafted packets, or repeated connection attempts to multiple ports can trigger alerts.
- Log analysis of firewall denies, unusual ARP traffic (Netdiscover), and endpoint monitoring for tools such as CurrPorts or Zenmap can identify reconnaissance early.

### Defensive Controls:

- Implement network segmentation, host-based firewalls, and intrusion-detection systems that flag or block scanning patterns.
- Disable unnecessary services, use port knocking or non-standard ports, and enable logging for all inbound connection attempts.
- Deploy honeypots, rate-limiting on routers, and continuous network monitoring to detect and respond to reconnaissance attempts.
- Apply least-privilege principles and regular vulnerability scanning from the defensive side to reduce the attack surface revealed by reconnaissance.

### Completion Verification:

- Work was completed across conceptual scanning review, Microsoft-side banner grabbing and GUI scanning, Linux-side Nmap/NSE scans, passive discovery, and advanced packet crafting.
- Completion was evidenced by live host lists from multiple tools, detailed Nmap output showing open ports/services/OS, banner-grab results, and connection-monitoring artifacts.
- The actual commands, scan profiles, and target ranges used in the lab are preserved directly in this report for accurate documentation and reproducibility.

### Key Findings:

- 1) Network scanning is the foundation of reconnaissance and reveals live hosts, open ports, running services, and OS details before any exploitation begins.
- 2) Both GUI (Zenmap, MyLanViewer) and CLI (Nmap, Hping3) tools provide complementary visibility into the same network environment.
- 3) Stealth scanning techniques (SYN, NULL, FIN, Xmas) and passive discovery (Netdiscover) allow reconnaissance while minimizing detection risk.
- 4) Banner grabbing and service enumeration reveal detailed software versions and OS fingerprints that feed directly into vulnerability exploitation.
- 5) The strongest defensive takeaway was that early detection of scanning activity through logs, IDS/IPS, and network monitoring is critical to preventing targeted attacks.

### Issues / Deviations / Notes:

- The lab used deprecated software and a legacy environment, so exact menu appearance and behavior could vary.
- Some tools required manual installation from the Intranet Hacking Tools page (e.g., ID Serve, Zenmap, CurrPorts, MyLanViewer).
- CurrPorts connection-closure required elevated privileges, which was noted as an expected limitation in the sandbox.

### Reflection:

- This lab tied together host discovery, port scanning, service enumeration, banner grabbing, and stealth techniques into a complete reconnaissance workflow across both Windows and Linux toolsets.
- The strongest takeaway was that scanning is dangerous because it bridges passive network visibility and targeted exploitation, making both detection controls and network hardening essential.